

University of Cape Town
University Examinations – May/June 2024
Department of Mathematics and Applied Mathematics
MAM3000W

Module 3MS: Metric Spaces

Duration : 2 hours

Full Marks: 40

Marks available: 42

1. Let X be a non-empty set and $f : X \rightarrow X$ an injective (i.e one-to-one) function.
Let d be a metric on X . Define p as follows, for $x, y \in X$,

$$p(x, y) = d(f(x), f(y)).$$

- (a) Prove that p is a metric on X . State clearly where the condition that f is injective is used.
- (b) Now let $X = \mathbb{R}$, $d = d_E$ (the usual metric), and $f(x) = e^x$. Define metric p as above and find the open ball $B(0, 4)$ in $(\mathbb{R}, p)$. (No proof required.) [10]
2. Prove: Let (X, d) be a metric space and A a subset of X , then $\text{bd}(A) = \emptyset$ if and only if A is both closed and open in X . (Here $\text{bd}(A)$ denotes the boundary of A as always). [6]
3. (a) Prove or disprove:
Let (X, d) be a metric space. If A and B are subsets of X , then $\overline{A \cup B} = \overline{A} \cup \overline{B}$.
- (b) Prove or disprove:
Let (X, d) be a metric space and A a finite subset of X , then A is complete.
- (c) Prove:
Let (X, d) be a metric space and T be a two-element set equipped with the discrete metric, then (X, d) is connected if and only if every continuous function $f : X \rightarrow T$ is constant. [12]

P.T.O

4. Read this result, and answer the questions that follow:

Proposition If the metric space (X, d) is compact, then this condition holds:
For any decreasing sequence (A_n) of non-empty subsets of X ,

$$\bigcap_{n \in \mathbb{N}} \overline{A_n} \neq \emptyset.$$

Proof Suppose (A_n) is a decreasing sequence of non-empty subsets of X . (1)

For each $n \in \mathbb{N}$, take $a_n \in A_n$. (2)

There exists a subsequence (a_{n_k}) of (a_n) such that $a_{n_k} \rightarrow x$ for some $x \in X$. (3)

We show that $x \in \bigcap_{n \in \mathbb{N}} \overline{A_n}$. (4)

Let $m \in \mathbb{N}$; we show $x \in \overline{A_m}$. (5)

Suppose $r > 0$. (6)

Take $N \in \mathbb{N}$ such that $k \geq N$ implies that $a_{n_k} \in B(x; r)$. (7)

Take $K \in \mathbb{N}$ such that $K \geq N$ and $n_K > m$. (8)

Then $B(x; r) \cap A_m \neq \emptyset$. (9)

(a) What is a “decreasing” sequence of subsets, as referred to in line (1)?

(b) How do we know that the subsequence mentioned in line (3) exists?

(c) How do we know the N mentioned in line (7) exists?

(d) How do we know the K mentioned in line (8) exists?

(e) Explain how line (9) follows from the previous working.

[10]

5. Recall that an ultrametric space is a metric space (X, d) in which the following strengthened form of the triangle inequality holds: for any $x, y, z \in X$,

$$d(x, z) \leq \max\{d(x, y), d(y, z)\}.$$

Prove that a metric space is ultrametric if and only if $\text{diam } B(x, r) \leq r$ for every open ball $B(x, r)$ in the space. [4]

THE END